Diamond Price Prediction

A Machine Learning Regression Project

53,794

Diamond Records

9

Predictors

2

ML Models Built

98.07%

Best Accuracy

Supervised Regression | Python | Scikit-learn

Project Overview

Problem Definition & ML Approach

🎯 Objective

Predict the price of a diamond (in USD) based on its physical and quality characteristics using supervised ML regression.

📊 Dataset

53,940 raw diamonds with 10 features. After removing 146 duplicates, 53,794 clean records were used for modelling.

🧠 ML Type

Supervised Learning — Regression, because the target variable (Price) is continuous and labelled data is available.

Project Workflow

- Data Loading & Deduplication
- Log Transformation of Target
- Outlier Treatment
- Missing Value Imputation
- Exploratory Data Analysis
- Feature Selection (Correlation + ANOVA)
- Encoding Categorical Variables
- Train-Test Split (70/30)
- Linear Regression & Decision Tree
- Cross-Validation & Accuracy Report

Dataset Description

10 features spanning physical dimensions and quality grades

price
Target

Price in USD ($326 – $18,823)

carat
Continuous

Weight of diamond (0.2 – 5.0 ct)

cut
Categorical

Quality: Fair → Good → Very Good → Premium → Ideal

color
Categorical

Grade J (worst) to D (best) — 7 levels

clarity
Categorical

Clarity: I1 → SI2 → SI1 → VS2 → VS1 → VVS2 → VVS1 → IF

depth
Continuous

Total depth % (z / mean(x,y))

table
Continuous

Width of top facet relative to widest point

x / y / z
Continuous

Length, width, and depth in mm

Data Preprocessing

Cleaning and transforming data for reliable modelling

01 Duplicate Removal

Dataset originally had 53,940 rows.
146 exact duplicate rows removed → 53,794 clean records retained.

02 Log Transformation of Target

Price was positively skewed — not suitable for regression.
log(Price) applied → bell-shaped distribution achieved.

03 Outlier Treatment

x: Values below 3.73 replaced with 3.73 (nearest logical value).
y: Values above 20 replaced with 10.54.
z: Values above 8 replaced with 6.98.

04 Missing Value Imputation

color: 6 missing → filled with Mode ('G').
depth: 14 missing → filled with Median (61.8).
All columns 100% complete after treatment.

Exploratory Data Analysis

Visual analysis of categorical and continuous variables

Categorical Variables — Bar Chart Distribution

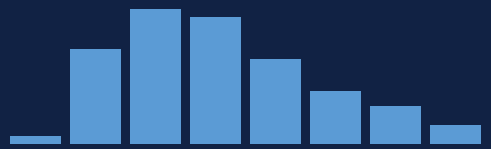
Cut

Fair Good V.Good Premium Ideal

Color

D E F G H I J

Clarity



I1 SI2 SI1 VS2 VS1 VVS2 VVS1 IF

✓ *All 3 categorical variables show sufficient frequency across categories — selected for modelling.*

Continuous Variables — Histogram Insights

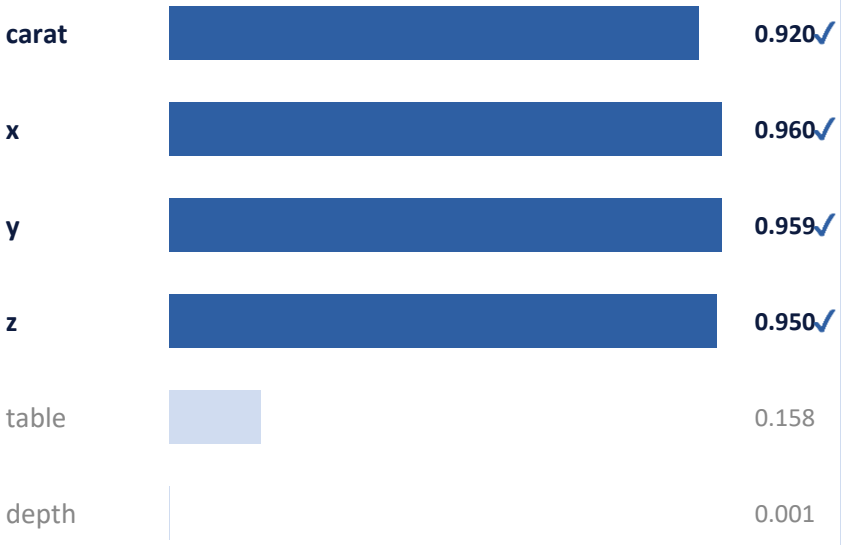
price (raw)	log(price)	carat	depth	x / y / z
⚠ Skewed	✓ Normal	✓ Selected	✓ Selected	✓ Treated
Positively skewed — unsuitable	Bell-shaped after log transform	Right-skewed — acceptable	Near-normal distribution	Outliers capped, improved shape

Feature Selection

Statistical methods to identify meaningful predictors

Pearson Correlation with log(Price)

Threshold: |r| > 0.5



Selected: carat, x, y, z

ANOVA Test — Categorical Predictors

H_0: No relationship between predictor & log(Price)
Reject H_0 if p-value < 0.05

CUT

p-value: 1.18e-208

✓ **Correlated with log(Price)**

COLOR

p-value: 1.87e-298

✓ **Correlated with log(Price)**

CLARITY

p-value: < 1e-300

✓ **Correlated with log(Price)**

All 3 categorical variables selected!

Final Feature Set & Encoding

7 predictors selected — categorical variables label-encoded

Feature	Type	Encoding	Role
carat	Continuous	None (numeric)	Predictor
cut	Categorical	Good=1, V.Good=2, Fair=3, Ideal=4, Premium=5	Predictor
color	Categorical	J=1 (worst) → D=7 (best)	Predictor
clarity	Categorical	I1=1 → IF=8 (best)	Predictor
x	Continuous	None (numeric)	Predictor
y	Continuous	None (numeric)	Predictor
z	Continuous	None (numeric)	Predictor
log_Price	**Continuous**	**log(price) transformation**	**Target ★**

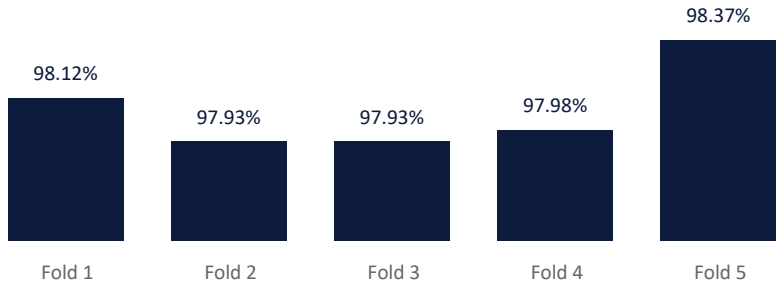
Train-Test Split: 70% Training (37,655 rows) | 30% Testing (16,139 rows) | random_state = 31

Model 1: Multiple Linear Regression

Baseline regression model on the cleaned & encoded dataset

0.9630	96.03%	96.36%	98.07%
R^2 (Training)	Mean Accuracy (Test)	Median Accuracy (Test)	Cross-Val Accuracy (10-Fold)

10-Fold Cross-Validation Accuracy



Avg: 98.07%

Key Insights

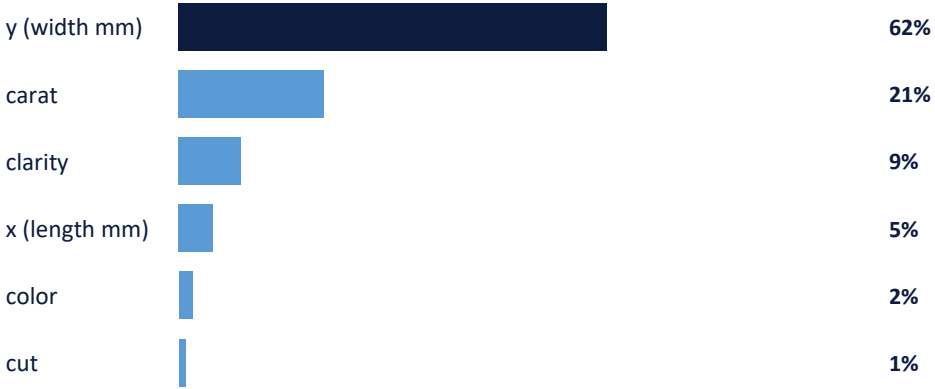
- R^2 of 0.963 on training data — very high explanatory power
- Test accuracy of 96% demonstrates strong generalizability
- Cross-validated accuracy of 98.07% confirms model robustness
- log transformation was key to satisfying linearity assumptions
- Low variance across 10 folds shows consistent performance

Model 2: Decision Tree Regressor

max_depth = 5 | criterion = friedman_mse

0.9613	95.81%	96.25%	96.83%
R^2 (Training)	Mean Accuracy (Test)	Median Accuracy (Test)	Cross-Val Accuracy (10-Fold)

Feature Importance (Top Predictors)



How the Tree Works

Root Split:	y ≤ 5.605mm separates low vs high price
max_depth = 5:	Tree limited to 5 levels to prevent overfitting
friedman_mse:	Improved Friedman MSE used as split criterion
y dominates:	Diamond width (y) is the strongest price predictor
Interpretable:	Clear decision rules readable by non-technical stakeholders

Model Comparison & Key Takeaways

Summary of findings and recommendations

Metric	Linear Regression	Decision Tree
R^2 (Training)	**0.9630**	0.9613
Mean Test Accuracy	**96.03%**	95.81%
Median Test Accuracy	**96.36%**	96.25%
10-Fold CV Accuracy	**98.07%**	96.83%
Interpretability	Moderate	High (rules visible)
Risk of Overfitting	Low	Medium (depth-limited)

★ **WINNER: Linear Regression**

- Log transformation was critical to achieving regression-ready target distribution
- All categorical features (cut, color, clarity) are statistically significant via ANOVA
- carat, x, y, z are the strongest continuous price predictors ($r > 0.92$)
- Linear Regression achieves 98.07% CV accuracy — recommended for production use

What This Project Proves

Skills and competencies demonstrated through end-to-end model development

Data Wrangling

146 duplicates removed · outliers capped · 20 NaNs imputed

Ability to audit raw data, apply domain-informed outlier capping, and impute missing values using median/mode — keeping data analysis-ready without leakage.

Statistical Thinking

Pearson $r > 0.92$ · ANOVA $p < 0.05$ on all 3 categoricals

Goes beyond visual EDA — uses Pearson correlation and ANOVA to statistically justify every feature selected, replacing guesswork with evidence.

Model Building & Evaluation

$R^2 = 0.963$ · 10-fold CV accuracy = 98.07%

Trains, tests, and cross-validates two ML algorithms; interprets R^2 and MAPE; and selects the winning model using rigorous, reproducible evidence.

Business Awareness

log(Price) transformation · ordinal label encoding for cut, color, clarity

Applying log transformation for skewness and encoding quality grades with their natural rank reflects real-world domain thinking — not just textbook ML.

Core Skills Demonstrated

Python	pandas · numpy · sklearn · scipy
EDA	histograms · boxplots · scatter plots
Statistics	Pearson correlation · ANOVA
ML Models	Linear Regression · Decision Tree
Validation	10-fold Cross-Validation · MAPE
Encoding	Label encoding · get_dummies()
Preprocessing	Outlier capping · median/mode imputation

★ End-to-end ownership: from raw CSV to a cross-validated, deployment-ready regression model achieving 98.07% accuracy